

Discrete Mathematics

CS1111

Lecture 3

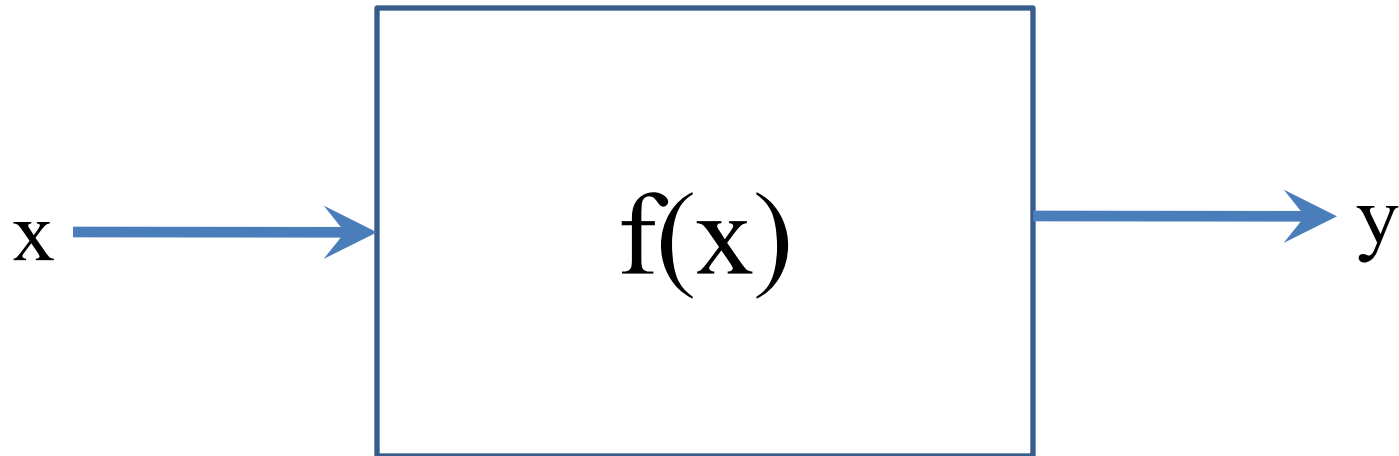
Functions

Functions

- A function is a relation with two conditions
 - All elements of the domain have images in the co-domain
 - No element in the domain relates to more than one element in the co-domain
- ➔ every member of the domain has exactly one image in the co-domain

Functions

- If f is a function from X to Y we write $f:X \rightarrow Y$, for every $x \in X$ there is an ordered pair $(x, y) \in f$. we write $y = f(x)$ “ y equals f of x ”
- For every input $x \in X$, there is a $y \in Y$



Functions

- Let A and B be nonempty sets. A function f from A to B is an assignment of exactly one element of B to each element of A .
- We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A .
- If f is a function from A to B , we write

$$f:A \rightarrow B$$

Functions

- If f is a function from A to B , we say that A is the domain of f and B is the co-domain of f .
- If $f(a) = b$, we say that b is the image of a and a is a pre-image of b .
- The range, or image, of f is the set of all images of elements of A . Also, if f is a function from A to B , we say that f maps A to B .
- A function is also called a mapping or transformation

Example

- Let $A = \{1, 2, 3\}$ and $B = \{a, b, c, d, e\}$.
Consider the relation
- $g \subseteq A \times B : g = \{(1, d), (2, c), (3, c)\}$
- This is a function – each element of A is paired with a single element of B (both 2 and 3 are paired with c but that's allowed).

Example

- By contrast $h = \{(1, a), (3, c), (1, d)\}$
- is not a function, because 1 is paired with two things, and 2 isn't paired with anything.
- Exercise : Which of the following relations from $\{a, b, c\} \times \{1, 2, 3\}$ are functions:
 - $R_1 = \{(a, 1), (a, 2), (b, 3), (c, 2)\}$
 - $R_2 = \{(a, 1), (b, 2), (c, 1)\}$
 - $R_3 = \{(a, 1), (c, 2)\}$

Functions and formulae

- It is actually rare to specify a function as a list of pairs. More usually, we give a rule or formula that makes it possible to work out the value (output) of a function on any given element of its domain(input).
- For instance, we can define a function $f \subseteq \mathbf{R} \times \mathbf{R}$ by the formula: $f(x) = x^2 + 3$

Functions and formulae

- To work out the value of the function we just substitute in the formula:

$$f(4) = 4^2 + 3 = 16 + 3 = 19$$

- or in another way of writing the same thing $(4, 19) \in f$. We could also write:

$$f = \{(x, x^2 + 3) \mid x \in \mathbf{R}\}$$

Exercise

- Determine the domain, co-domain and range of each of the following functions
 - $f: \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(x) = x^2$ for all values of $x \in \mathbb{Z}$
 - $g: \mathbb{N} \rightarrow \mathbb{N}$ such that $g(x) = 2x$
 - $h: \mathbb{N} \rightarrow \mathbb{N}$ such that $h(x) = 2x + 1$

One to One Functions

- A function $f : A \rightarrow B$ is said to be a one to one (1: 1) or injective if for every element $b \in B$ there is at most one element $a \in A$ such that $f(a) = b$
- That is for every $x, y \in A$ if $x \neq y$ then $f(x) \neq f(y)$

Or

- Every element in the co-domain B is an image for only one element in the domain A .

One to One Functions

- Example: Determine if each of the following functions is 1 : 1 or not
 - $f: \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(x) = x^2$ for all values of $x \in \mathbb{Z}$
 - $f(-a) = f(a)$ hence it is not one-one
 - $g: \mathbb{N} \rightarrow \mathbb{N}$ such that $g(x) = 2x + 3$
 - If $a \neq b$ then $g(a) \neq g(b)$ hence it is one-one

Onto Functions

- A function $f: A \rightarrow B$ is said to be onto or surjective if for every element $b \in B$ there is an element $a \in A$ such that $f(a) = b$
- All elements of the co-domain are images for elements in the domain.
- $\Rightarrow$ the range = co-domain

Onto Functions

- Example: Determine if each of the following functions is onto or not
 - $f: \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(x) = x^2$ for all values of $x \in \mathbb{Z}$
 - There is no $x \in \mathbb{Z}$ such that $f(x) = -1$. All members of $\mathbb{Z}^-$ are not members of the range of f . hence it is not onto
 - $g: \mathbb{N} \rightarrow \mathbb{N}$ such that $g(x) = 2x + 3$
 - We cannot find an $x \in \mathbb{N}$ such that $g(x) = 10$. $\rightarrow$ range $\neq$ co-domain. Hence it is not onto.

One to One Correspondence

- A function $f(x)$ is said to one-one correspondence or bijective if it is one-one and onto.

Exercises

- For each of the following functions, show whether it is one-one, onto or both (bijective)
 - $f_1(x) = x^3$
 - $f_2(x) = e^x$ Exponential function
 - $f_3(x) = ax + b$ Linear function
 - $f_4(x) = x^2$ Quadratic function
 - $f_5(x) = x$ Identity function
 - $f_6(x) = a$ Constant function

Inverse Function

- Let $f(x) : X \rightarrow Y = \{(x, y) \mid x \in X \text{ and } y \in Y\}$

The inverse of a function $f(x)$ denoted $f^{-1}(x)$ is a relation such that $f^{-1}(x): Y \rightarrow X = \{(y, x) \mid y = f(x)\}$

- If $f(x)$ is bijective then $f^{-1}(x)$ is a function otherwise it is not a function.
- Example
 - Let $f(x) = x^2$ be a function on the set of reals. Then $f^{-1}(x) = \sqrt{x}$, $f^{-1}(25) = \pm 5 \rightarrow 25$ has two images $\rightarrow f^{-1}(x)$ is not a function.
 - $f(x) = x^2$ is neither one-one nor onto.

Composition of Functions

- Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$, the composition of f and g denoted $f \circ g$ is defined as

$$f \circ g(x) = f(g(x))$$

- Example: let $f(x) = 3x + 2$, and $g(x) = x^2$.

$$f \circ g(x) = f(g(x)) = f(x^2) = 3x^2 + 2$$

$$g \circ f(x) = g(f(x)) = g(3x + 2) = (3x + 2)^2 = 9x^2 + 12x + 4$$

- Exercise: find $f \circ g(3)$ and $g \circ f(2)$

Composition of Functions

- Note

- $f \circ g(x) \neq g \circ f(x)$

- ➔ NOT commutative

- $f \circ (g \circ h) = (f \circ g) \circ h$

- ➔ associative